
How to look up the unique PCR-RFLP genotype # of an allelic profile

Introduction

For *Toxoplasma gondii*, the common multilocus genotyping approach uses a panel of markers such as **SAG1, SAG2, SAG3, BTUB, GRA6, c22-8, c29-2, L358, PK1, and Apico**. Historically, these markers have been typed using **PCR-RFLP**, in which allele types are assigned based on differences in restriction-enzyme digestion patterns.

These days, an alternative **in silico approach** called **multilocus sequence typing (MLST)** can be used when sequence data is available. Sequences from the marker loci are analyzed computationally to determine the allele at each locus directly and generate a multilocus genotype profile for each isolate.

This tutorial describes how to use an isolate's multilocus genotype profile to determine its corresponding unique PCR-RFLP genotype number.

Example

Your sample may have the following multilocus genotype profile

Sample01

SAG1	5' + 3' SAG2	Alt. SAG2	SAG3	BTUB	GRA6	C22-8	C29-2	L358	PK1	APICO
I	I	I	III	I	II	u-1	I	I	III	III

Step 1: Access the **genotypes table** at the link below.

This table has been sourced from Dr. Chunlei Su, University of Tennessee Knoxville, and has been provided on ToxoDB as a community resource.

https://beta.toxodb.org/toxo.beta/app/workspace/analyses/EDAUD_BN_R5Y9Zx5R0FA/new/variables/ENT_275a7000/VAR_84064fd6

The screenshot shows the ToxoDB Dataset Explorer interface. At the top, there's a search bar and navigation links. The main section is titled "Dataset Explorer" and "Table of unique PCR-RFLP genotypes". Below this, there's a "Records" box showing "354 of 354". The interface is divided into a left sidebar with a list of loci and a main content area. The main content area shows the "SAG1 genotype" variable, with a table of records and a distribution bar chart. The table has columns for "Subset of Records", "All Records", and "Distribution". The distribution bar chart shows the frequency of alleles: "I" (246, 70%), "II or III" (47, 13%), "u-1" (56, 16%), and "u-2" (1, < 1%).

Notice the list of loci on the left. When you click on each locus name, you will see a list of alleles on the right.

Step 2: Go through the list and **for each locus, click the allele type that your sample corresponds to**. Repeat for all loci.

For instance-

- For **SAG1 genotype**, click "I"

This screenshot shows the "SAG1 genotype" variable selected in the left sidebar. The main content area displays the "SAG1 genotype" variable with a table of records. The table has columns for "Subset of Records", "All Records", and "Distribution". The distribution bar chart shows the frequency of alleles: "I" (246, 70%), "II or III" (47, 13%), "u-1" (56, 16%), and "u-2" (1, < 1%). The "I" allele is selected in the table, and the "Keep checked values at top" option is checked.

- For the **5' + 3' SAG2 genotype**, click "I"

This screenshot shows the "5' + 3' SAG2 genotype" variable selected in the left sidebar. The main content area displays the "5' + 3' SAG2 genotype" variable with a table of records. The table has columns for "Subset of Records", "All Records", and "Distribution". The distribution bar chart shows the frequency of alleles: "I" (246, 70%), "II or III" (47, 13%), "u-1" (56, 16%), and "u-2" (1, < 1%). The "I" allele is selected in the table, and the "Keep checked values at top" option is checked.

Step 3: Finally, you're at the **PCR-RFLP genotype** variable.

Do three steps as indicated in the figure below:

1. Click on the PCR-RFLP Genotype ID variable in the list at the left
2. For the Subset of Records columns in the table, click on the ordering button twice to sort it in descending order
3. Note the PCR-RFLP genotype # that fits your allele selection

You are filtering the data on your unique allele combination and the relevant genotype # will be apparent at the top of the list.

The screenshot shows the iVISTA software interface. At the top, a 'Records' box indicates '1 of 354'. Below this, a series of tabs represent different variables: SAG1 genotype, 5' + 3' SAG2 genotype, Alt. SAG2 genotype, SAG3 genotype, BTUB genotype, GRA6 genotype, C22-8 genotype, C29-2 genotype, and L358 genotype. The 'Browse and Subset' tab is active. On the left, a 'Record' list shows various variables, with 'PCR RFLP Genotype ID' selected. A red arrow points to this selection with the annotation '(1) click on the PCR-RFLP Genotype ID variable'. The main panel displays the 'PCR RFLP Genotype ID' variable details, including its original name 'PCR_RFLP_Genotype' and a description: 'Unique identifier number assigned to each distinct multi-locus PCR-RFLP genotype in the dataset. A few numbers are not assigned'. Below this, a table shows the subset of records. The table has columns for 'PCR RFLP Genotype ID', 'Subset of Records', and a percentage. The first row shows genotype #81 with 1 record (100%). A red arrow points to the 'Subset of Records' column header with the annotation '(2) click on the subset ordering button TWICE'. Another red arrow points to the value '#81' in the first row with the annotation '(3) Note the PCR RFLP Genotype ID number for your multi-locus allele set'. The table lists genotype IDs from #81 down to #92, all with 0 records (0%).

PCR RFLP Genotype ID	Subset of Records	Percentage
#81	1	(100%)
#99	0	(0%)
#98	0	(0%)
#97	0	(0%)
#96	0	(0%)
#95	0	(0%)
#94	0	(0%)
#93	0	(0%)
#92	0	(0%)

You're done! You have identified the unique PCR-RFLP genotype # for the allele profile of your sample.